Examiner only

2. *(a)* State the difference between *baryons* and *mesons* in terms of quark make-up. [2]

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(b) When two protons collide, the following interaction may occur, where x is an unknown particle:

$$\mathrm{p} + \mathrm{p} \longrightarrow \mathrm{p} + x + \pi^0$$

(i) The π^0 is a meson which carries no charge. State its quark make-up. [1]

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(ii) Identify particle x, explaining how you use the law of conservation of baryon number and one other conservation law. [3]

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(iii) State how lepton number is conserved in the above interaction. [1]

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Examiner only

(c) A π^0 decays in a typical time of 8×10^{-17} s into two photons as shown. State which force is involved in this interaction, giving **two** reasons for your answer. [3]

$$\pi^0 \longrightarrow \gamma + \gamma$$

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